

machines and equipments in the fields ranging from agriculture to automobiles, infotainment to health, processing to construction industry and toys to aircrafts. Halappa says, "I would encourage fresh/experienced engineers to consider this field as one of the top options as it offers highly challenging, learning and rewarding career for engineers with enthusiasm to be part of this futuristic industry."

"Mobile devices are driving the electronics development. Hence, the designs are miniaturised and have several challenges to be addressed. The engineers need to strengthen their knowledge on currently used devices, which are not taught in colleges, on their own by studying and experimenting with kits available in market," informs Patel. Halappa advises, "There are lot of low-cost development kits available in market to play with new processors and software architectures. Most of them share schematics, design files and firmware on the Internet, which can be a good way to learn embedded system design aspects to get entry in organisations."

Patel says, "There is lack of good-quality staff and infrastructure in many colleges. Knowledge of faculty in current technologies is also very limited. Therefore our advice to people looking for career in embedded systems will be to attend short-term courses in embedded systems or take an entry-level position in an organisation that specialises in embedded product design at low salary instead of taking up QA/testing/application development job at higher salary in traditional IT companies." On a similar note, Tiwari says, "Students need to invest their time in learning hands-on practical skills and concepts which are currently in demand. They should try to become industry-ready and only then they can get a job of choice in this market."

"Candidates need to develop right

skills and attitude. Apart from the technical knowledge and consistent academic performance, companies are in need of people having passion towards technology, who keep them abreast with the latest advancements in the technology, who take initiatives in their field and add value to their work," opines VEDI.

Can engineering institutes be blamed?

Nowadays, you get to hear quite often that engineering graduates are raw and unemployable. The blame, conveniently, is transferred to the institutions and the government, feels VEDI. She says, "I admit that it is the onus of the institutions and government to bring a change in the course curriculum and give an ideal environment to young students where they can learn and adopt necessary skills helpful for them to be employed in the industry." However, she adds, "I feel that students should take initiative and get involved in proactive measures to learn and develop skills which would help them in the long run. Internet and social media, groups and communities have made the information accessible to them in just one click. They should spend their time on enhancing their knowledge plus also work on improving their communication skills. After all, what companies require from a fresh graduate is just the basics plus good communication and, above all, right attitude."

Halappa agrees that the industry often complains that fresh graduates, coming out of regular institutions, are not ready for the industry. "It does not mean that the curriculum in these institutions is inferior, but in fact it is

somewhat over-ambitious and looks like super popular packages of all the latest technologies tailored to please every company that is coming for campus recruitment. So the real need is to ensure that the engineering colleges concentrate more on teaching better basics and providing practical orientation to contemporary subjects than dealing with every advanced and research area," he says.

Halappa believes that subjects need to be dealt in top-down approach rather than bottom-up approach. He cites, "The top-down approach would first give overall perspective of the subject, even before getting into the details of it. Unfortunately, the current and more prevalent bottom-up approach allows students to learn bits and pieces of the subject that may be easier to learn and score even without getting the overall perspective of the subject."

For this change to happen, Halappa feels the text books should be written with top-down approach. The teaching fraternity in the engineering institutions, who are the key players in this eco system, should be allowed, encouraged and even compelled to interact with industries on regular basis and gain insight to the product design life-cycle, design methodologies, standards and compliance, and the kind of job profiles available to engineers in the industry. "Only when a comprehensive system for training the trainer is established by the responsible and committed involvement of both industry and institutions, this gap can be minimised," according to Halappa. ●

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